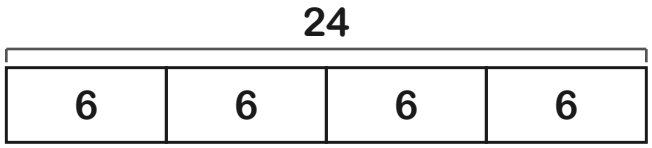
# All Four Operations

Two-step problems with +, −, × and ÷. Draw the bar.

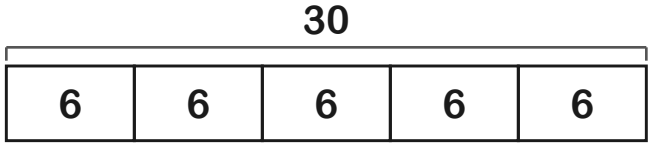
### 4 bags with 6 apples in each: equal pieces




$4 \times 6 = 24$

Know the pieces? Multiply.

### 30 stickers shared by 5 kids

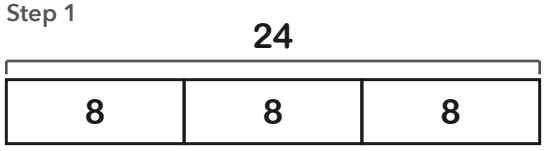



$30 \div 5 = 6$

Know the whole? Divide.

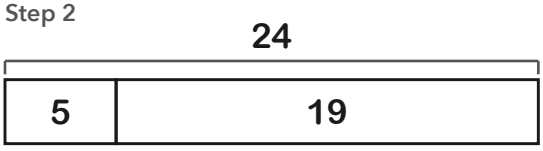
### Two steps: 3 packs of 8 pencils. 5 get lost. How many are left?

Step 1




$3 \times 8 = 24$

Step 2



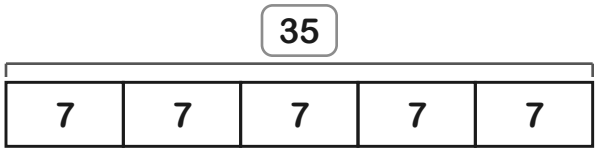

$24 - 5 = 19$

### Does it make sense? Round to check.

$38 + 47 = 85$ $\quad$ about $40 + 50 = 90$

85 is close to 90, so it makes sense. 58 would not.

**Your turn!**




$\square \times \square = \square$ (5 × 7 = 35)

**For teachers**

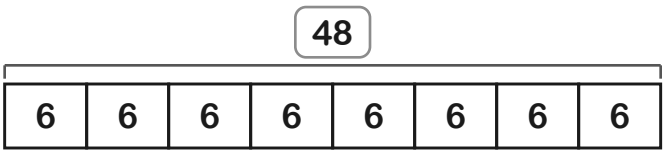
Goal: two-step problems with +, −, × and ÷ (3.OA.8). Equal pieces mean × (whole missing) or ÷ (a piece missing). Solve step 1, then use it in step 2. Round to check the answer makes sense.

# All Four Operations: Practice

Read. Draw the bar. Write the number sentence. Answer.

**Part 1** Equal pieces: find the whole Every piece is the same. Multiply.

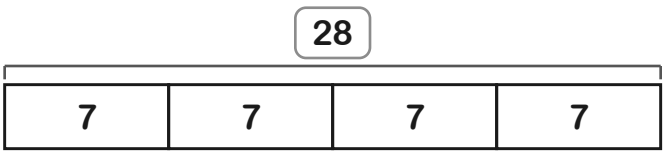
1) Tia buys 8 boxes of crayons. Each box has 6 crayons. How many crayons does she have?




Number sentence 8 × 6 = 48

Answer **48** crayons

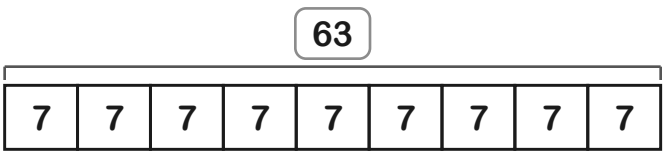
2) There are 4 tables. 7 kids sit at each table. How many kids are there?




Number sentence 4 × 7 = 28

Answer **28** kids

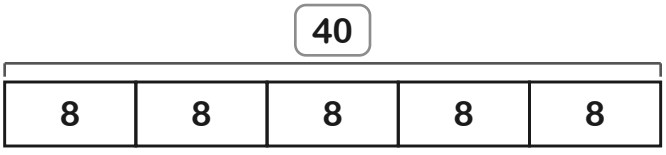
3) There are 9 vases with 7 flowers in each vase. How many flowers is that?




Number sentence 9 × 7 = 63

Answer **63** flowers

4) Each pack has 8 juice boxes. Ms. Lee buys 5 packs. How many juice boxes is that?



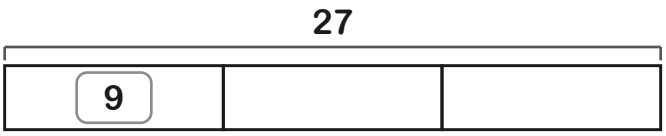

Number sentence 5 × 8 = 40

Answer **40** juice boxes

## All Four Operations: Practice (continued)

**Part 2** Equal pieces: find one piece The whole is shared equally. Divide.

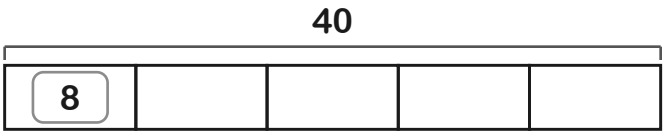
5) 27 kids make 3 equal teams. How many kids are on each team?




Number sentence **27 ÷ 3 = 9**

Answer **9** kids

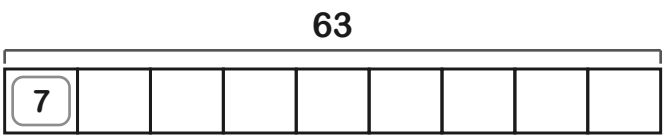
6) A ribbon 40 inches long is cut into 5 equal pieces. How long is each piece?




Number sentence **40 ÷ 5 = 8**

Answer **8** inches

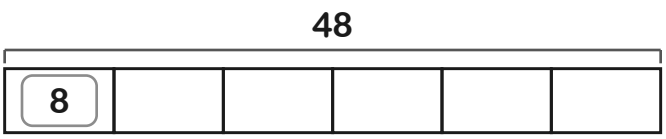
7) 63 apples are packed into 9 equal bags. How many apples are in each bag?




Number sentence **63 ÷ 9 = 7**

Answer **7** apples

8) Dad shares 48 grapes equally among 6 kids. How many grapes does each kid get?




Number sentence **48 ÷ 6 = 8**

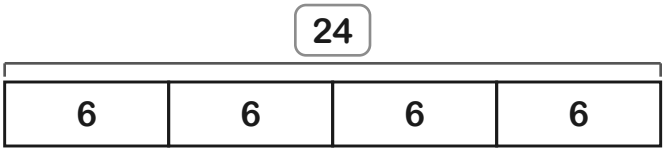
Answer **8** grapes

## All Four Operations: Practice (continued)

**Part 3** Two steps The bars are drawn for you. Step 1 helps with step 2.

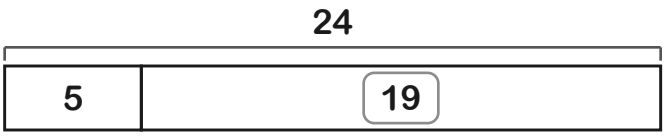
9) Ben buys 4 packs of 6 cookies. He eats 5 cookies. How many cookies are left?

Step 1




Step 1 $4 \times 6 = 24$

Step 2

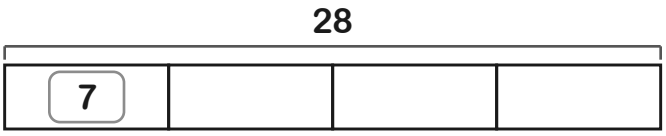



Step 2 $24 - 5 = 19$

Answer **19** cookies

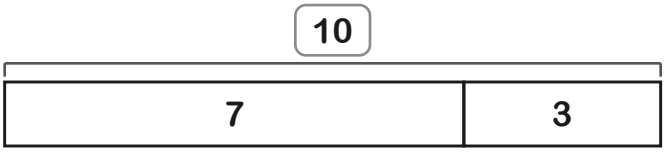
10) 28 kids sit at 4 tables, the same number at each. Then 3 more kids join each table. How many kids are at each table now?

Step 1




Step 1 $28 \div 4 = 7$

Step 2




Step 2 $7 + 3 = 10$

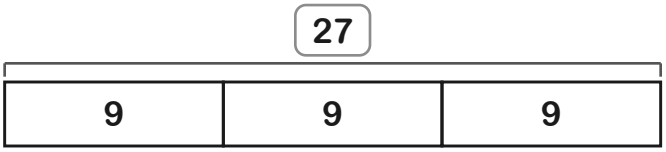
Answer **10** kids

## All Four Operations: Practice (continued)

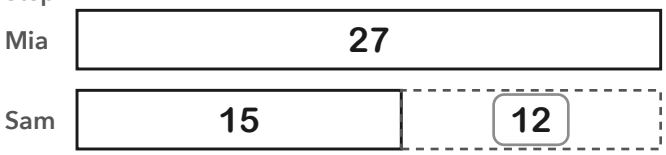
**Part 3** Two steps (continued)

11) Mia has 3 bags of 9 marbles. Sam has 15 marbles. How many more marbles does Mia have?

Step 1




Step 2




Step 1 3 × 9 = 27

Step 2 15 + 12 = 27

Answer 12 marbles

**Part 4** Two steps on your own Draw a bar for each step.

12) A class has 5 rows of 6 desks. 8 desks are empty. How many desks have kids at them?

Draw the bars

Step 1 5 × 6 = 30

Step 2 30 − 8 = 22

Answer 22 desks

## All Four Operations: Practice (continued)

**Part 4** Two steps on your own (continued)

13) Kim shares 45 beads equally among 5 friends. Each friend uses 3 beads. How many beads does each friend have left?

Draw the bars

Step 1 **45 ÷ 5 = 9**

Step 2 **9 − 3 = 6**

Answer **6** beads

14) Stickers cost $4 a pack. Leo buys 6 packs and pays with $30. How much change does he get?

Draw the bars

Step 1 **6 × $4 = $24**

Step 2 **$30 − $24 = $6**

Answer **$6**

## All Four Operations: Practice (continued)

**Part 4** Two steps on your own (continued)

15) A farmer has 7 rows of 8 corn plants and 20 tomato plants. How many plants are there in all?

Draw the bars

Step 1 $7 \times 8 = 56$

Step 2 $56 + 20 = 76$

Answer **76** plants

**Part 5** Does it make sense? Round each number to the nearest ten. About how much?

16) $38 + 47$ is about **90**

17) $24 + 33$ is about **50**

18) $82 - 37$ is about **40**

19) $56 + 41$ is about **100**

20) $93 - 21$ is about **70**

21) $66 + 12$ is about **80**

22) $58 - 33$ is about **30**

23) $71 - 49$ is about **20**

## All Four Operations: Practice (continued)

**Part 6** More stories Times or divide? Draw a bar if it helps.

24) Kai shares 28 books equally among 4 friends. How many books does each friend get?

**28 ÷ 4 = 7**

Answer **7** books

25) Ella shares 16 blocks equally among 4 friends. How many blocks does each friend get?

**16 ÷ 4 = 4**

Answer **4** blocks

26) Leo shares 24 pencils equally among 3 friends. How many pencils does each friend get?

**24 ÷ 3 = 8**

Answer **8** pencils

27) Ravi shares 32 stickers equally among 8 friends. How many stickers does each friend get?

**32 ÷ 8 = 4**

Answer **4** stickers

28) Kim has 7 bags with 7 crayons in each bag. How many crayons are there in all?

**7 × 7 = 49**

Answer **49** crayons

29) Lily has 3 bags with 7 leaves in each bag. How many leaves are there in all?

**3 × 7 = 21**

Answer **21** leaves

30) Leo has 3 bags with 5 toy cars in each bag. How many toy cars are there in all?

**3 × 5 = 15**

Answer **15** toy cars

## All Four Operations: Practice (continued)

**Part 6** More stories (continued)

31) Ben shares 35 leaves equally among 7 friends. How many leaves does each friend get?

**35 ÷ 7 = 5**

Answer **5** leaves

32) Kai shares 42 cards equally among 6 friends. How many cards does each friend get?

**42 ÷ 6 = 7**

Answer **7** cards

33) Theo shares 24 toy cars equally among 6 friends. How many toy cars does each friend get?

**24 ÷ 6 = 4**

Answer **4** toy cars

34) Nia has 5 bags with 3 rocks in each bag. How many rocks are there in all?

**5 × 3 = 15**

Answer **15** rocks

35) Omar shares 21 toy cars equally among 7 friends. How many toy cars does each friend get?

**21 ÷ 7 = 3**

Answer **3** toy cars

## All Four Operations: Practice (continued)

**Part 7** Remember? Problems from earlier packets.

36) $3 \times 80 =$ **240**

37) 6 cm by 9 cm: area = **54** sq cm

38) 7 m by 2 m: perimeter = **18** m

39) 20 minutes after 3:50 = **4:10**

40) Round 438 to the nearest hundred: **400**

41) 1 hour 30 minutes = **90** minutes

**Part 8** Word problems No help this time. Two steps each.

42) A box holds 6 eggs. Mom buys 5 boxes and uses 12 eggs. How many eggs are left?

Number sentence $5 \times 6 = 30, 30 - 12 = 18$

Answer **18** eggs

Draw or show your work

43) Ana has 32 stickers. She gives 4 friends 5 stickers each. How many stickers are left?

Number sentence $4 \times 5 = 20, 32 - 20 = 12$

Answer **12** stickers

Draw or show your work

44) 24 kids stand in 3 equal lines. 2 kids leave each line. How many kids are in each line now?

Number sentence $24 \div 3 = 8, 8 - 2 = 6$

Answer **6** kids

Draw or show your work

## All Four Operations: Practice (continued)

### Part 8 Word problems (continued)

45) Tickets cost $7 each. Ben buys 3 tickets and a $9 snack. How much does he spend?

Number sentence **3 × $7 = $21, $21 + $9 = $30**

Answer **$30**

Draw or show your work

46) A garden has 9 rows of 4 plants. Then 10 more plants are added. How many plants are there now?

Number sentence **9 × 4 = 36, 36 + 10 = 46**

Answer **46** plants

Draw or show your work

47) Challenge: Leo reads 8 pages a day for 7 days. His book has 100 pages. How many pages are left?

Number sentence **7 × 8 = 56, 100 − 56 = 44**

Answer **44** pages

Draw or show your work

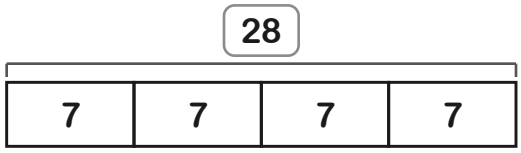
# Check-In: All Four Operations

Show what you know. No help this time.

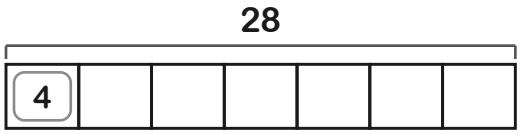
**Mastery: 5 of 6 (80%)**

**Part 1** Equal pieces Find the number that goes in the box.

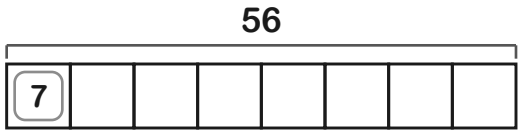
1)



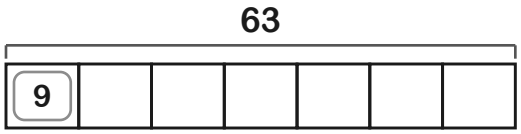

2)




3)



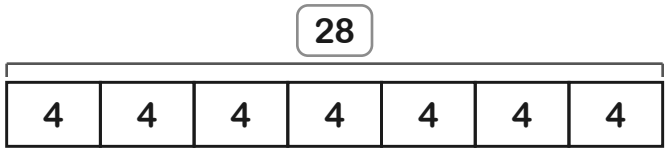

4)




**Part 2** Two steps

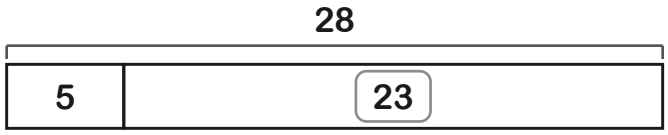
5) Lily has 7 bags with 4 books in each bag. Lily gives away 5 books. How many books are left?

Step 1




Step 1 $7 \times 4 = 28$

Step 2




Step 2 $28 - 5 = 23$

Answer **23** books

**Part 3** Two-step word problems

6) Ben has 5 boxes of 5 buttons. Ben uses 16 of them. How many buttons are left?

$5 \times 5 = 25, 25 - 16 = 9$

Answer **9** buttons